

Manual Change History (5411181) X15 CM2350 X114B - Efficiency Series Service Manual

Date	11-Nov-2016
Section	Back
Procedure Number	5411181-history (/qs3/pubsys2/xml/en/manual/5411181/5411181-history.html)
Procedure Title	Manual Change History
Update Reason	New manual for X15 CM2350 X114B released for use.

Date	05-Dec-2016
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM - No Communication Troubleshooting Tree
Update Reason	Modified Steps 2, 6, 9 and 11 and added information for new INLINE™ 7 data link.

Date	12-Dec-2016
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	00-019-472 (/qs3/pubsys2/xml/en/procedures/00/00-019-472.html)
Procedure Title	INSITE™ Communication Errors
Update Reason	Removed or corrected information for INLINE™ 4, 5, and 6. Added new information for INLINE™ 7 data link.

Date	13-Dec-2016
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Removed or corrected information for INLINE™ 4, 5, and 6. Added new information for INLINE™ 7 data link.
Date	17-Jan-2017
Section	Section TS - Troubleshooting Symptoms
Procedure Number	5411181-t092-tr (/qs3/pubsys2/xml/en/manual/5411181/5411181-t092-tr.html)
Procedure Title	Fuel in the Lubricating Oil
Update Reason	Added tree node pertaining to aftertreatment system active regeneration time.
Date	25-Jan-2017
Section	Section TS - Troubleshooting Symptoms
Procedure Number	5411181-t087-tr (/qs3/pubsys2/xml/en/manual/5411181/5411181-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	Added tree node pertaining to the time between DPF regenerations.
Date	03-Feb-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-121 (/qs3/pubsys2/xml/en/procedures/493/493-011-121.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Suction Test
Update Reason	Updated specification in the Measure Step.

Date	03-Feb-2017
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-023 (/qs3/pubsys2/xml/en/procedures/377/377-010-023.html)
Procedure Title	Air Intake Manifold
Update Reason	Corrected link to 010-131.
Date	03-Feb-2017
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-131 (/qs3/pubsys2/xml/en/procedures/377/377-010-131.html)
Procedure Title	Air Intake Connection Adapter
Update Reason	Added to outline.
Date	20-Jan-2017
Section	Section 1 - Cylinder Block
Procedure Number	377-001-033 (/qs3/pubsys2/xml/en/procedures/377/377-001-033.html)
Procedure Title	Gear Housing, Front
Update Reason	Added graphic and text to General Information to identify gear trains with a solid style lower idler gear.
Date	09-Feb-2017
Section	Section 1 - Cylinder Block
Procedure Number	377-001-055 (/qs3/pubsys2/xml/en/procedures/377/377-001-055.html)
Procedure Title	Gear Train Backlash, Front
Update Reason	Added service instructions for gear trains with a solid style lower idler gear.

Date	16-Feb-2017
Section	Section 1 - Cylinder Block
Procedure Number	377-001-088 (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
Procedure Title	Engine Base Timing
Update Reason	Added service instructions for gear trains with a solid style lower idler gear.
Date	08-Mar-2017
Section	Section 6 - Injectors and Fuel Lines - Group - 06
Procedure Number	377-006-021 (/qs3/pubsys2/xml/en/procedures/377/377-006-021.html)
Procedure Title	Fuel Manifold (Drain)
Update Reason	Updated to thermal recirculation valve (TRV) for field service events.
Date	08-Mar-2017
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-003 (/qs3/pubsys2/xml/en/procedures/377/377-006-003.html)
Procedure Title	Air in Fuel
Update Reason	Updated thermal recirculation valve (TRV) troubleshooting with longer bleed time.
Date	17-Mar-2017
Section	Section 2 - Cylinder Head
Procedure Number	377-002-014 (/qs3/pubsys2/xml/en/procedures/377/377-002-014.html)
Procedure Title	Injector Sleeve, Cylinder Head
Update Reason	Added procedure to manual.

Date	03-Apr-2017
Section	Section TS - Troubleshooting Symptoms
Procedure Number	5411181-t091-tr (/qs3/pubsys2/xml/en/manual/5411181/5411181-t091-tr.html)
Procedure Title	Fuel in Coolant
Update Reason	Updated "Cause" to match "Correction" in first node.
Date	14-Mar-2017
Section	Section TS - Troubleshooting Symptoms
Procedure Number	5411181-t256-tr (/qs3/pubsys2/xml/en/manual/5411181/5411181-t256-tr.html)
Procedure Title	Connected Calibrations Malfunctioning
Update Reason	Revised and added t256 to manual.
Date	29-Mar-2017
Section	Section 11 - Exhaust System
Procedure Number	377-011-031 (/qs3/pubsys2/xml/en/procedures/377/377-011-031.html)
Procedure Title	EGR Cooler Coolant Lines
Update Reason	Updated EGR cooler graphics.
Date	04-Apr-2017
Section	Section 9 - Drive Units
Procedure Number	377-009-056-tr (/qs3/pubsys2/xml/en/procedures/377/377-009-056-tr.html)
Procedure Title	Air Leaks, Air Intake and Exhaust Systems
Update Reason	Bullet added for inspecting for loose clamps or damage (Air compressor intake tube).

Date	06-Apr-2017
Section	Section 9 - Drive Units
Procedure Number	377-009-056 (/qs3/pubsys2/xml/en/procedures/377/377-009-056.html)
Procedure Title	Charging System Alternator/Refrigerant Compressor Mounting Bracket
Update Reason	Revise graphics for X15.
Date	19-Apr-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-019 (/qs3/pubsys2/xml/en/procedures/377/377-011-019.html)
Procedure Title	EGR Cooler
Update Reason	Updated graphics in Remove and Install steps; also updated text regarding removal of EGR cooler mounting nuts on the lubricating oil cooler assembly.
Date	19-Apr-2017
Section	Section 9 - Drive Units
Procedure Number	377-012-109 (/qs3/pubsys2/xml/en/procedures/377/377-012-109.html)
Procedure Title	Air Compressor Inlet Tube
Update Reason	Added graphics for X15.
Date	19-Apr-2017
Section	Section 12 - Compressed Air System
Procedure Number	377-012-109 (/qs3/pubsys2/xml/en/procedures/377/377-012-109.html)
Procedure Title	Air Compressor Inlet Tube
Update Reason	Added, Naturally aspirated air compressor adapter to air compressor inlet tube.

Date	20-Apr-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-036 (/qs3/pubsys2/xml/en/procedures/493/493-011-036.html)
Procedure Title	Aftertreatment Selective Catalytic Reduction (SCR) Catalyst
Update Reason	Updated graphic in Install step; also updated text regarding orientation of drain hole location when installing a new SCR catalyst.
Date	04-May-2017
Section	Section 5 - Fuel System - Group 5
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Updated the chart listed for the Fuel Pump Performance Test.
Date	11-May-2017
Section	Section 14 - Engine Testing - Group 14
Procedure Number	493-014-025 (/qs3/pubsys2/xml/en/procedures/493/493-014-025.html)
Procedure Title	Aftertreatment Selective Catalytic Reduction (SCR) System Test
Update Reason	Removed "INSITE™ electronic service tool, version 8.1.1 or later" from System Requirements; also updated ECM calibration release date.
Date	31-May-2017
Section	Section 11 - Exhaust System
Procedure Number	00-011-151 (/qs3/pubsys2/xml/en/procedures/00/00-011-151.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Pressure Relief Valve
Update Reason	Added to manual.

Date	05-May-2017
Section	Section 11 - Exhaust System
Procedure Number	493-011-121 (/qs3/pubsys2/xml/en/procedures/493/493-011-121.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Suction Test
Update Reason	Updated dosing unit vacuum specification.
Date	13-Jun-2017
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-227 (/qs3/pubsys2/xml/en/procedures/377/377-005-227.html)
Procedure Title	Fuel Pump Head(s)
Update Reason	Added Rebuild step and video.
Date	29-Jun-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-086 (/qs3/pubsys2/xml/en/procedures/493/493-011-086.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Line Restriction Test
Update Reason	Updated dosing unit intake restriction specifications.
Date	18-Jul-2017
Section	Section 14 - Engine Testing - Group 14
Procedure Number	493-014-025 (/qs3/pubsys2/xml/en/procedures/493/493-014-025.html)
Procedure Title	Aftertreatment Selective Catalytic Reduction (SCR) System Test
Update Reason	Updated the System Requirements step.

Date	28-Jul-2017
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-033 (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
Procedure Title	Turbocharger
Update Reason	Updated procedure to remove potential misunderstandings about replacing turbochargers.

Date	28-Jul-2017
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Removed 'excessive force' statement.

Date	22-Aug-2017
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	493-019-478 (/qs3/pubsys2/xml/en/procedures/493/493-019-478.html)
Procedure Title	Aftertreatment Particulate Sensor
Update Reason	Updated Test information, Connect service tool to "aftertreatment particulate sensor" instead of "NOx sensor module."

Date	30-Aug-2017
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	377-006-081 (/qs3/pubsys2/xml/en/procedures/377/377-006-081.html)
Procedure Title	Fuel System Pressure Relief Procedure
Update Reason	Removed fuel rail nut torque stepblock.

Date	11-Sep-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-121 (/qs3/pubsys2/xml/en/procedures/493/493-011-121.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Suction Test
Update Reason	Updated line art.
Date	28-Sep-2017
Section	Section 5 - Fuel System
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Added a Minimum Fuel Volume Measurement During High Pressure Leakage Test table.
Date	09-Oct-2017
Section	Section 6 - Injectors and Fuel Lines - Group 6
Procedure Number	337-006-060 (/qs3/pubsys2/xml/en/procedures/337/337-006-060.html)
Procedure Title	Fuel Rail
Update Reason	Updated Install Step; Fuel lines must be loosened to relieve potential stress prior to tightening fuel rail capscrews.
Date	11-Oct-2017
Section	Section 2 - Clinder Head
Procedure Number	377-002-014 (/qs3/pubsys2/xml/en/procedures/377/377-002-014.html)
Procedure Title	Injector Sleeve, Cylinder Head
Update Reason	Added Clean and Inspect for Reuse step.

Date	22-Aug-2017
Section	Section L - Service Literature
Procedure Number	459-205-001 (/qs3/pubsys2/xml/en/procedures/459/459-205-001.html)
Procedure Title	Additional Service Literature
Update Reason	Added Service Bulletin 5467581.
Date	22-Aug-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-036 (/qs3/pubsys2/xml/en/procedures/493/493-011-036.html)
Procedure Title	Aftertreatment Selective Catalytic Reduction (SCR) Catalyst
Update Reason	Revised the Clean and Inspect for Reuse step to reference Service Bulletin 5467581.
Date	31-Aug-2017
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	493-002-004 (/qs3/pubsys2/xml/en/procedures/493/493-002-004.html)
Procedure Title	Cylinder Head
Update Reason	Updated procedure with revised part number for valve vacuum tester.
Date	27-Oct-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-121 (/qs3/pubsys2/xml/en/procedures/493/493-011-121.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Suction Test
Update Reason	Revised the Measure step to include removal of the DEF dosing unit inlet screen filter.

Date	15-Nov-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-080 (/qs3/pubsys2/xml/en/procedures/493/493-011-080.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid System Leak Test
Update Reason	Removed purging information mentioned in the dosing fluid leak test.

Date	28-Nov-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-059 (/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Valve
Update Reason	Added resistance measurement specifications.

Date	07-Dec-2017
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-645 (/qs3/pubsys2/xml/en/procedures/377/377-019-645.html)
Procedure Title	Route Parameter Manager Module
Update Reason	Replaced references of Advanced Dynamic Efficient Powertrain Technology (ADEPT™) support electronic service tool with reference to Cummins Guidanz™ electronic service tool.

Date	07-Dec-2017
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-647 (/qs3/pubsys2/xml/en/procedures/377/377-019-647.html)
Procedure Title	Route Parameter Manager Module Calibration Code
Update Reason	Replaced reference of Advanced Dynamic Efficient Powertrain Technology (ADEPT™) support electronic service tool with reference to Cummins Guidanz™ electronic service tool.

Date	07-Dec-2017
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-648 (/qs3/pubsys2/xml/en/procedures/377/377-019-648.html)
Procedure Title	Route Parameter Manager Module Global Positioning System (GPS) Map
Update Reason	Replaced reference of Advanced Dynamic Efficient Powertrain Technology (ADEPT™) support electronic service tool with reference to Cummins Guidanz™ electronic service tool.
Date	20-Nov-2017
Section	Section 10 - Air Intake System
Procedure Number	377-010-033 (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
Procedure Title	Turbocharger
Update Reason	Revised Initial step to include service bulletin and troubleshooting trees.
Date	13-Dec-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-047 (/qs3/pubsys2/xml/en/procedures/493/493-011-047.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Differential Pressure Sensor Tubes
Update Reason	Removed hyperlinks.
Date	13-Dec-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-056 (/qs3/pubsys2/xml/en/procedures/493/493-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Removed hyperlinks.

Date	13-Dec-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-058 (/qs3/pubsys2/xml/en/procedures/493/493-011-058.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit
Update Reason	Updated tool part number.
Date	13-Dec-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-059 (/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Valve
Update Reason	Added hyperlinks.
Date	13-Dec-2017
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-085 (/qs3/pubsys2/xml/en/procedures/493/493-011-085.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Restriction Test
Update Reason	Removed hyperlinks.
Date	13-Dec-2017
Section	Section 14 - Engine Testing - Group 14
Procedure Number	493-014-016 (/qs3/pubsys2/xml/en/procedures/493/493-014-016.html)
Procedure Title	Aftertreatment Diesel Particulate Filter (DPF) Regeneration Test
Update Reason	Removed hyperlinks.

Date	13-Dec-2017
Section	Section 14 - Engine Testing - Group 14
Procedure Number	493-014-017 (/qs3/pubsys2/xml/en/procedures/493/493-014-017.html)
Procedure Title	Snap Acceleration Test
Update Reason	Removed hyperlinks.
Date	14-Dec-2017
Section	Section 6 - Injectors and Fuel Lines - Group 6
Procedure Number	377-006-012 (/qs3/pubsys2/xml/en/procedures/377/377-006-012.html)
Procedure Title	Fuel Drain Line Restriction
Update Reason	Revised graphic in Setup step.
Date	15-Dec-2017
Section	Section TS - Troubleshooting Symptoms
Procedure Number	5411181-t128-tr (/qs3/pubsys2/xml/en/manual/5411181/5411181-t128-tr.html)
Procedure Title	Diesel Exhaust Fluid Usage - Abnormal
Update Reason	Updated to match EDS.
Date	02-Jan-2018
Section	Section 10 - Air Intake System
Procedure Number	377-010-150 (/qs3/pubsys2/xml/en/procedures/377/377-010-150.html)
Procedure Title	Turbocharger Actuator Calibration Code
Update Reason	Removed bullet statement from General Information to Install.

Date	12-Jan-2018
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-144 (/qs3/pubsys2/xml/en/procedures/493/493-011-144.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing System Pressure Test
Update Reason	Added pressure tool connection points and normal pressure value table.

Date	19-Jan-2018
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-033 (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
Procedure Title	Turbocharger
Update Reason	Corrected Leak Test for coolant loop on X15 engines.

Date	19-Jan-2018
Section	Section TS - Troubleshooting Symptoms
Procedure Number	5411181-t122-tr (/qs3/pubsys2/xml/en/manual/5411181/5411181-t122-tr.html)
Procedure Title	Turbocharger Leaks Engine Oil or Fuel
Update Reason	Updated TS tree t122 to reference TS tree T185 and T186 troubleshooting trees.

Date	22-Jan-2018
Section	Section 11 - Exhaust System
Procedure Number	377-011-024 (/qs3/pubsys2/xml/en/procedures/377/377-011-024.html)
Procedure Title	EGR Cooler Connection
Update Reason	Removed, thermostat housing from the Remove step.

Date	12-Feb-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-055 (/qs3/pubsys2/xml/en/procedures/377/377-001-055.html)
Procedure Title	Gear Train Backlash, Front
Update Reason	Updated gear backlash measurements.
Date	13-Feb-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-088 (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
Procedure Title	Engine Base Timing
Update Reason	Updated entire procedure.
Date	20-Feb-2018
Section	Section TS - Troubleshooting Symptoms
Procedure Number	5411181-t044-tr (/qs3/pubsys2/xml/en/manual/5411181/5411181-t044-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (No Exhaust Smoke)
Update Reason	Changed wording from gear pump to fuel pump gear pump.
Date	28-Feb-2018
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-059 (/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Valve
Update Reason	Changed the gasket to a ring seal.

Date	28-Feb-2018
Section	Section F - Familiarization
Procedure Number	377-011-999 (/qs3/pubsys2/xml/en/procedures/377/377-011-999.html)
Procedure Title	Exhaust System - Overview
Update Reason	Removed paragraph about DEF dosing unit defrosting.

Date	20-Mar-2018
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-016 (/qs3/pubsys2/xml/en/procedures/377/377-005-016.html)
Procedure Title	Fuel Pump
Update Reason	Fuel pump timing improvements were added to reduce gear-train noise.

Date	05-Apr-2018
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Updated the 'Maximum Injector Return Flow' measurement.

Date	10-Apr-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-079 (/qs3/pubsys2/xml/en/procedures/377/377-001-079.html)
Procedure Title	Gear Cover, Upper
Update Reason	Updated graphics.

Date	09-May-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-017 (/qs3/pubsys2/xml/en/procedures/377/377-001-017.html)
Procedure Title	Crankshaft Adapter
Update Reason	Updated incorrect crankshaft adapter drill hole specifications in the Remove step while updating the procedure to current standards.
Date	01-Jun-2018
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	377-003-011-tr (/qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html)
Procedure Title	Rocker Lever Cover
Update Reason	Provided clear instruction to the field as to where vibration isolator capscrews should be assembled in the rocker lever cover.
Date	27-Jun-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	377-007-031 (/qs3/pubsys2/xml/en/procedures/377/377-007-031.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Updated spring compressor part number and spring bore depth specification.
Date	03-Jul-2018
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-043 (/qs3/pubsys2/xml/en/procedures/377/377-011-043.html)
Procedure Title	Aftertreatment Adapter Pipe
Update Reason	Revised 'NOx sensor module' to 'NOx sensor probe.'

Date	03-Jul-2018
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-463 (/qs3/pubsys2/xml/en/procedures/377/377-019-463.html)
Procedure Title	Aftertreatment Intake NOx Sensor
Update Reason	Revised 'NOx sensor module' to 'NOx sensor probe.'

Date	10-Jul-2018
Section	Section 10 - Air Intake System
Procedure Number	377-010-150 (/qs3/pubsys2/xml/en/procedures/377/377-010-150.html)
Procedure Title	Turbocharger Actuator Calibration Code
Update Reason	Changed graphic and added bullet to clarify when to key off.

Date	23-Jul-2018
Section	Section F - Familiarization
Procedure Number	377-200-003 (/qs3/pubsys2/xml/en/procedures/377/377-200-003.html)
Procedure Title	Flow Diagram, Cooling System
Update Reason	Updated the rear out turbocharger flow diagram.

Date	27-Jul-2018
Section	Section 7 - Lubricating Oil System
Procedure Number	377-007-003 (/qs3/pubsys2/xml/en/procedures/377/377-007-003.html)
Procedure Title	Lubricating Oil Cooler
Update Reason	The leak test was changed to a pressure decay test with an acceptable leak rate.

Date	02-Aug-2018
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-665 (/qs3/pubsys2/xml/en/procedures/377/377-019-665.html)
Procedure Title	Advanced Dynamic Efficient Powertrain Technology (ADEPT™)
Update Reason	Added procedure to the manual.
Date	30-Aug-2018
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-390 (/qs3/pubsys2/xml/en/procedures/377/377-019-390.html)
Procedure Title	Turbocharger Speed Sensor
Update Reason	Removed twisting verbiage in Remove section.
Date	06-Aug-2018
Section	Section 7 - Lubricating Oil System
Procedure Number	377-007-031 (/qs3/pubsys2/xml/en/procedures/377/377-007-031.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Revised graphic showing arrangement of valve, spring, and cap.
Date	07-Sept-2018
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Revised graphics in Install step to include guide pins.

Date	15-Oct-2018
Section	Section 11 - Exhaust System
Procedure Number	493-011-060 (/qs3/pubsys2/xml/en/procedures/493/493-011-060.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Filter
Update Reason	Added, Lubricate the o-ring prior to installation.

Date	18-Oct-2018
Section	Section 5 - Fuel System
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Fuel Rail Return Flow check if there is no fuel flow during the High-Pressure Fuel Pump Performance Test initial priming steps added.

Date	30-Oct-2018
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-020 (/qs3/pubsys2/xml/en/procedures/377/377-006-020.html)
Procedure Title	Fuel Inlet Restriction
Update Reason	Revised Select Service Tools and graphics since it utilizes the quick connect instead of a banjo fitting.

Date	16-Jan-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-088 (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
Procedure Title	Engine Base Timing
Update Reason	Updated torque value for lower idler gears.

Date	17-Jan-2019
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	377-003-004-tr (/qs3/pubsys2/xml/en/procedures/377/377-003-004-tr.html)
Procedure Title	Overhead Set
Update Reason	Updated graphics.
Date	12-Feb-2019
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	493-019-449 (/qs3/pubsys2/xml/en/procedures/493/493-019-449.html)
Procedure Title	Aftertreatment Exhaust Gas Temperature Sensor
Update Reason	Revised graphics in General Information step.
Date	18-Feb-2019
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	377-007-013 (/qs3/pubsys2/xml/en/procedures/377/377-007-013.html)
Procedure Title	Lubricating Oil Filter (Spin-On)
Update Reason	Replaced graphics in General Information step.
Date	14-Feb-2019
Section	Section V - Sepecifications
Procedure Number	377-018-017 (/qs3/pubsys2/xml/en/procedures/377/377-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Replaced Full Flow / Bypass with NanoNet® Filter Specifications.

Date	18-Feb-2019
Section	Section V - Sepecifications
Procedure Number	377-018-024 (/qs3/pubsys2/xml/en/procedures/377/377-018-024.html)
Procedure Title	Cummins®/Fleetguard® Filter Specifications
Update Reason	Revised Part Numbers for the Lubricating Oil Filter.

Date	08-Apr-2019
Section	Section 19 - Electronic Controls
Procedure Number	377-019-397 (/qs3/pubsys2/xml/en/procedures/377/377-019-397.html)
Procedure Title	Fuel Lift Pump Circuit
Update Reason	Updated procedure to include connector required for voltage check during Key-On.

Date	18-Apr-2019
Section	Section 3 - Rocker Levers
Procedure Number	377-003-004 (/qs3/pubsys2/xml/en/procedures/377/377-003-004.html)
Procedure Title	Overhead Set
Update Reason	Corrected the minimum value for the intake valve and revised graphic in General Information.

Date	18-Apr-2019
Section	Section 3 - Rocker Lever Assembly
Procedure Number	377-003-009 (/qs3/pubsys2/xml/en/procedures/377/377-003-009.html)
Procedure Title	Rocker Lever Assembly
Update Reason	Updated graphic to show clockwise direction.

Date	18-Apr-2019
Section	Section 2 - Cylinder Head
Procedure Number	377-002-024 (/qs3/pubsys2/xml/en/procedures/377/377-002-024.html)
Procedure Title	Overhead Camshaft, Injector
Update Reason	Update graphic to show housing/engine of a single camshaft.

Date	18-Apr-2019
Section	Section 2 - Cylinder Head
Procedure Number	377-002-016 (/qs3/pubsys2/xml/en/procedures/377/377-002-016.html)
Procedure Title	Valve Guide Seal, Cylinder Head
Update Reason	Updated Preparatory/Finishing Steps to remove/install the rocker lever shaft assemblies adjust the overhead set as needed.

Date	22-Apr-2019
Section	Section E - Engine and System Identification
Procedure Number	377-100-002 (/qs3/pubsys2/xml/en/procedures/377/377-100-002.html)
Procedure Title	Engine Diagrams
Update Reason	Reversed components air intake connection and crankcase breather housing for number 11 and 12 on Left Side View.

Date	06-May-2019
Section	Seciton 1 - Cylinder Block - Group 01
Procedure Number	377-001-088 (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
Procedure Title	Engine Base Timing
Update Reason	Added a stepblock at the end of the Inspect for Reuse section to inspect the upper idler gear shaft assembly.

Date	06-May-2019
Section	Seciton 1 - Cylinder Block - Group 01
Procedure Number	377-001-111 (/qs3/pubsys2/xml/en/procedures/377/377-001-111.html)
Procedure Title	Idler Gear, Upper
Update Reason	Added Disassemble and Assemble steps.
Date	3-May-2019
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	377-016-007 (/qs3/pubsys2/xml/en/procedures/377/377-016-007.html)
Procedure Title	Flywheel Housing, REPTO
Update Reason	Modified rear end power take-off (REPTO) backlash limits.
Date	07-May-2019
Section	Section 1 - Cylinder Block
Procedure Number	377-001-033 (/qs3/pubsys2/xml/en/procedures/377/377-001-033.html)
Procedure Title	Gear Housing, Front
Update Reason	Added Procedure 002-050 to the Preparatory and Finishing Steps.
Date	08-May-2019
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	377-002-050 (/qs3/pubsys2/xml/en/procedures/377/377-002-050.html)
Procedure Title	Camshaft Retainer Plate
Update Reason	Added to outline.
Date	08-May-2019
Section	Section 6 - Injectors and Fuel Lines - Group 6
Procedure Number	377-006-012 (/qs3/pubsys2/xml/en/procedures/377/377-006-012.html)
Procedure Title	Fuel Drain Line Restriction
Update Reason	Revised graphic in Setup step.

Date	11-Jun-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-121 (/qs3/pubsys2/xml/en/procedures/493/493-011-121.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Suction Test
Update Reason	Rearraged procedure to correct misinformation and ease confusion from the field.

Date	11-Jun-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-143 (/qs3/pubsys2/xml/en/procedures/493/493-011-143.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Manual Purge
Update Reason	Removed procedure references for 011-080.

Date	12-Jun-2019
Section	Section 11 - Exhaust System
Procedure Number	493-011-086 (/qs3/pubsys2/xml/en/procedures/493/493-011-086.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Line Restriction Test
Update Reason	Graphics revised to show open position of valve.

Date	13-Jun-2019
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	377-007-025 (/qs3/pubsys2/xml/en/procedures/377/377-007-025.html)
Procedure Title	Lubricating Oil Pan
Update Reason	Created stepblock specifically for the oil drain plug.

Date	13-Jun-2019
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	377-002-004 (/qs3/pubsys2/xml/en/procedures/377/377-002-004.html)
Procedure Title	Cylinder Head
Update Reason	Added a note to the beginning of the Inspect for Reuse step.
Date	18-Jun-2019
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-020 (/qs3/pubsys2/xml/en/procedures/377/377-006-020.html)
Procedure Title	Fuel Inlet Restriction
Update Reason	Changed the graphic in the service procedure to a correct one and adding fuel inlet restriction test measurement in engine will not start section.
Date	20-Jun-2019
Section	Section 19 - Electronic Controls
Procedure Number	377-019-665 (/qs3/pubsys2/xml/en/procedures/377/377-019-665.html)
Procedure Title	Advanced Dynamic Efficient Powertrain Technology (ADEPT™)
Update Reason	Revised to match the INSITE™ information.
Date	11-Jul-2019
Section	Section 5 - Fuel System - Group 5
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Updated graphic's and fuel pump return flow test is modified with correct tools.

Date	15-Jul-2019
Section	Section 1 - Cylinder Block- Group 01
Procedure Number	377-001-058 (/qs3/pubsys2/xml/en/procedures/377/377-001-058.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Updated procedure to inform the field about brass shims.

Date	01-Aug-2019
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-027 (/qs3/pubsys2/xml/en/procedures/377/377-014-027.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Regeneration Analyzer
Update Reason	Added to manual.

Date	12-Aug-2019
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	377-006-003 (/qs3/pubsys2/xml/en/procedures/377/377-006-003.html)
Procedure Title	Air in Fuel
Update Reason	Update new design drain manifold.

Date	14-Aug-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-165 (/qs3/pubsys2/xml/en/procedures/493/493-011-165.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Voltage Drop Test
Update Reason	Added procedure to manual.

Date	16-Aug-2019
Section	Section 1 - Cylinder Block
Procedure Number	377-001-028 (/qs3/pubsys2/xml/en/procedures/377/377-001-028.html)
Procedure Title	Cylinder Liner
Update Reason	Modify the procedure when resurfacing the mid-stop counterbore.
Date	28-Aug-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-007 (/qs3/pubsys2/xml/en/procedures/377/377-011-007.html)
Procedure Title	Exhaust Manifold, Dry
Update Reason	Changed order of inspection process on the manifold and turbocharger flange. Overhead Camshaft, Injector
Date	17-Sep-2019
Section	Section 2 - Cylinder Head
Procedure Number	377-002-024 (/qs3/pubsys2/xml/en/procedures/377/377-002-024.html)
Procedure Title	Overhead Camshaft, Valve
Update Reason	Added Camshaft Removal and Installation Kit, Part Number 5395348 instructions.
Date	18-Sep-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-056 (/qs3/pubsys2/xml/en/procedures/493/493-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Updated the concentration of DEF percentage.

Date	20-Sep-2019
Section	Section 19 - Electrical Controls - Group 19
Procedure Number	377-019-397 (/qs3/pubsys2/xml/en/procedures/377/377-019-397.html)
Procedure Title	Fuel Lift Pump Circuit
Update Reason	Revised procedure to disregard checking the voltage after removing the fuel pump power connector.

Date	24-Sep-2019
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	377-006-012 (/qs3/pubsys2/xml/en/procedures/377/377-006-012.html)
Procedure Title	Fuel Drain Line Restriction
Update Reason	Update new design drain manifold.

Date	24-Sep-2019
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	377-006-013 (/qs3/pubsys2/xml/en/procedures/377/377-006-013.html)
Procedure Title	Fuel Drain Lines
Update Reason	Update new design drain manifold.

Date	24-Sep-2019
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	377-006-017 (/qs3/pubsys2/xml/en/procedures/377/377-006-017.html)
Procedure Title	Fuel Filter Head
Update Reason	Update new design drain manifold.

Date	24-Sep-2019
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	377-006-021 (/qs3/pubsys2/xml/en/procedures/377/377-006-021.html)
Procedure Title	Fuel Manifole (Drain)
Update Reason	Update new design drain manifold.

Date	27-Aug-2019
Section	Section 2 - Cylinder Head
Procedure Number	377-022-001_02 (/qs3/pubsys2/xml/en/procedures/377/377-022-001_02.html)
Procedure Title	Service Tools
Update Reason	Added Camshaft Removal and Installation Kit, Part Number 5395348.
Date	07-Oct-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-056 (/qs3/pubsys2/xml/en/procedures/493/493-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Updated with regeneration specification for a healthy engine.
Date	09-Oct-2019
Section	Section 7 - Lubricating Oil System
Procedure Number	377-007-003 (/qs3/pubsys2/xml/en/procedures/377/377-007-003.html)
Procedure Title	Lubricating Oil Cooler
Update Reason	Corrected the minimum depth specification for the install depth of the thermostat insert.
Date	10-Oct-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-062 (/qs3/pubsys2/xml/en/procedures/493/493-011-062.html)
Procedure Title	Aftertreatment Decomposition Tube
Update Reason	Replaced multi-module screen graphic with single-module graphic.

Date	14-Oct-2019
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-016 (/qs3/pubsys2/xml/en/procedures/377/377-005-016.html)
Procedure Title	Fuel Pump
Update Reason	Added information about freight train noise and timing the fuel pump with a new service tool.

Date	15-Oct-2019
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-665 (/qs3/pubsys2/xml/en/procedures/377/377-019-665.html)
Procedure Title	Advanced Dynamic Efficient Powertrain Technology (ADEPT™)
Update Reason	Added the features of predictive engine braking, on-ramp boost, and predictive cruise control to the procedure.

Date	21-Oct-2019
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	Engine Control Module Calibration Code
Update Reason	Added the following statement to the Preparatory Steps of the procedure: "Verify vehicle odometer is equal to the value recorded before ECM removal. Contact OEM service location if values are incorrect."

Date	21-Oct-2019
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-031 (/qs3/pubsys2/xml/en/procedures/377/377-019-031.html)
Procedure Title	Engine Control Module
Update Reason	Added the following statement to the Finishing Steps of the procedure: "Verify vehicle odometer is equal to the value recorded before ECM removal. Contact OEM service location if values are incorrect."
Date	15-Nov-2019
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	377-003-009 (/qs3/pubsys2/xml/en/procedures/377/377-003-009.html)
Procedure Title	Rocker Lever Assembly
Update Reason	Updated graphic 03c00249 to 03c00338.
Date	20-Dec-2019
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Added statement to Remove section and updated graphic. Added stepblock to Clean and Inspect for Reuse steps.
Date	02-Jan-2020
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Removed hose (Part Number 3164673) from the Select Service Tools list. Updated Banjo flow adapter fitting to Part Number 5299319.

Date	06-Jan-2020
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-152 (/qs3/pubsys2/xml/en/procedures/377/377-010-152.html)
Procedure Title	Variable Geometry Turbocharger Hysteresis Test
Update Reason	Added note to rerun the test if the turbocharger has been replaced.
Date	14-Jan-2020
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-012 (/qs3/pubsys2/xml/en/procedures/377/377-006-012.html)
Procedure Title	Fuel Drain Line Restriction
Update Reason	Added M22 Straight Thread O-Ring Plug (STOR) fitting, Part Number 3089238 to Select Service Tools list and Setup steps.
Date	14-Jan-2020
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-022-001_06 (/qs3/pubsys2/xml/en/procedures/377/377-022-001_06.html)
Procedure Title	Service Tools
Update Reason	Added M22 Straight Thread O-Ring Plug (STOR) fitting to Service Tools list.
Date	15-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-026 (/qs3/pubsys2/xml/en/procedures/377/377-001-026.html)
Procedure Title	Cylinder Block
Update Reason	Removed statement located in the Clean and Inspect for Reuse steps that called for all six cylinder liners to be replaced if fretting or wear was present.

Date	15-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-027 (/qs3/pubsys2/xml/en/procedures/377/377-001-027.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	Removed statement located in the Leak Test steps that called for all six cylinder liners to be replaced if fretting or wear was present.
Date	27-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-022-001_01 (/qs3/pubsys2/xml/en/procedures/377/377-022-001_01.html)
Procedure Title	Service Tools
Update Reason	Revised the listed part number for the marked socket from 3165161 to 5573072.
Date	27-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-006 (/qs3/pubsys2/xml/en/procedures/377/377-001-006.html)
Procedure Title	Bearings, Main
Update Reason	Revised the listed part number for the marked socket from 3165161 to 5573072.
Date	03-Feb-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-033 (/qs3/pubsys2/xml/en/procedures/377/377-001-033.html)
Procedure Title	Gear Housing, Front
Update Reason	Added caution statement to the final Clean and Inspect for Reuse step.

Date	04-Feb-2020
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-033 (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
Procedure Title	Turbocharger
Update Reason	Added service bulletin resources for damaged turbocharger compressor impellers and turbine wheels.

Date	06-Feb-2020
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-033 (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
Procedure Title	Turbocharger
Update Reason	Added note to Finishing Steps that indicates if a stationary aftertreatment regeneration is needed to complete an repair, the engine should be brought up to operating temperature before the regeneration has begun.

Date	10-Feb-2020
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-665 (/qs3/pubsys2/xml/en/procedures/377/377-019-665.html)
Procedure Title	Advanced Dynamic Efficient Powertrain Technology (ADEPT™)
Update Reason	Added information for the seventh feature, the Predictive Road Speed Governor.

Date	04-Mar-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-026 (/qs3/pubsys2/xml/en/procedures/377/377-014-026.html)
Procedure Title	Engine Cylinder Leak Down Test
Update Reason	Added correct part numbers into procedure.

Date	16-Mar-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-027 (/qs3/pubsys2/xml/en/procedures/377/377-014-027.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Regeneration Analyzer
Update Reason	Updated the procedure to require INSITE service tool version 8.5.2 or later in order to fully support ARA tests.
Date	25-Mar-2020
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	377-012-020 (/qs3/pubsys2/xml/en/procedures/377/377-012-020.html)
Procedure Title	Air Compressor (Oil Carryover)
Update Reason	Reconfigured procedure to commonize it.
Date	01-Apr-2020
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-417 (/qs3/pubsys2/xml/en/procedures/377/377-019-417.html)
Procedure Title	Data Link Circuit, Proprietary
Update Reason	Updated VDC specification to 1.0 throughout the procedure.
Date	14-Apr-2020
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Removed unnecessary warning from the Preparatory and Finishing Steps of the procedure.

Date	14-Apr-2020
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-041 (/qs3/pubsys2/xml/en/procedures/377/377-010-041.html)
Procedure Title	Turbocharger Coolant Hoses
Update Reason	Removed unnecessary warning from the Preparatory and Finishing Steps of the procedure.

Date	21-Apr-2020
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-060 (/qs3/pubsys2/xml/en/procedures/493/493-011-060.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Filter
Update Reason	Added instruction to lubricate the top foam ring of the filter housing before torquing the filter onto the pump housing.

Date	11-May-2020
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Corrected the listed part number for the high-pressure fuel pump service tool, from Part Number 5299754 to Part Number 5299753.

Date	11-May-2020
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-022-001_05 (/qs3/pubsys2/xml/en/procedures/377/377-022-001_05.html)
Procedure Title	Service Tools
Update Reason	Added Fuel Pump Test Kit, Part Number 5299721 to Select Service Tools list.

Date	15-May-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-058 (/qs3/pubsys2/xml/en/procedures/377/377-001-058.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Added Select Service Tools list to the procedure.
Date	19-May-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-022-001_01 (/qs3/pubsys2/xml/en/procedures/377/377-022-001_01.html)
Procedure Title	Service Tools
Update Reason	Added Counterbore Cutting Improvement Kit, Part Number 5395552 to Select Service Tools list.
Date	22-Jun-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-033 (/qs3/pubsys2/xml/en/procedures/377/377-001-033.html)
Procedure Title	Gear Housing, Front
Update Reason	Corrected incorrect procedure reference to point to the appropriate procedure.
Date	29-Jun-2020
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Corrected torque value listed in the Fuel Pressure Relief Valve Return Flow Test step.

Date	14-Jul-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-002 (/qs3/pubsys2/xml/en/procedures/377/377-014-002.html)
Procedure Title	Engine Testing (Chassis Dynamometer)
Update Reason	Added tri-axles and drive axles only.
Date	21-Jul-2020
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-059 (/qs3/pubsys2/xml/en/procedures/493/493-011-059.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Valve
Update Reason	Added new graphic to procedure to highlight the aftertreatment DEF signal pins.
Date	09-Jul-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-004 (/qs3/pubsys2/xml/en/procedures/377/377-014-004.html)
Procedure Title	Engine Run-in (Without Dynamometer)
Update Reason	Added procedure to manual.
Date	05-Aug-2020
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	493-019-443 (/qs3/pubsys2/xml/en/procedures/493/493-019-443.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Differential/Outlet Pressure Sensor
Update Reason	Updated initial check voltage specifications.

Date	07-Aug-2020
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-390 (/qs3/pubsys2/xml/en/procedures/377/377-019-390.html)
Procedure Title	Turbocharger Speed Sensor
Update Reason	Added instruction to Remove step to not discard mounting clips connectors. Added mention of turbocharger speed sensor into procedure. Corrected second graphic shown in Install steps.
Date	12-Aug-2020
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-026 (/qs3/pubsys2/xml/en/procedures/377/377-006-026.html)
Procedure Title	Injector
Update Reason	Added an updated o-ring installation graphic to the first stepblock of the Install steps.
Date	09-Nov-2020
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-041 (/qs3/pubsys2/xml/en/procedures/377/377-010-041.html)
Procedure Title	Turbocharger Coolant Hoses
Update Reason	Corrected torque value conversion.
Date	09-Nov-2020
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-056 (/qs3/pubsys2/xml/en/procedures/493/493-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Added note for technician to monitor specific parameters to determine why a stationary (parked) regeneration will not complete.

Date	18-Nov-2020
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-024-tr (/qs3/pubsys2/xml/en/procedures/377/377-010-024-tr.html)
Procedure Title	Air Leaks, Air Intake and Exhaust Systems
Update Reason	Reorganized procedure to include details pertaining to the smoke machine and service plug.
Date	15-Dec-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-047 (/qs3/pubsys2/xml/en/procedures/377/377-001-047.html)
Procedure Title	Piston Rings
Update Reason	Corrected minimum and maximum values listed for the Piston Ring Gap.
Date	15-Jan-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-052 (/qs3/pubsys2/xml/en/procedures/377/377-001-052.html)
Procedure Title	Vibration Damper, Viscous
Update Reason	Updated the graphic depicting the installation of the vibration damper, pulley and clamping plate.
Date	10-Feb-2021
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-021 (/qs3/pubsys2/xml/en/procedures/377/377-006-021.html)
Procedure Title	Fuel Manifold (Drain)
Update Reason	The graphics of the "install" and "assemble" steps were reversed in the procedure. The edit was to fix the order of the graphics to the appropriate order.

Date	12-Feb-2021
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-022-001_05 (/qs3/pubsys2/xml/en/procedures/377/377-022-001_05.html)
Procedure Title	Service Tools
Update Reason	Added Fuel Pump Timing Gauge, Part Number 5572664, to service tools list.

Date	12-Feb-2021
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-016 (/qs3/pubsys2/xml/en/procedures/377/377-005-016.html)
Procedure Title	Fuel Pump
Update Reason	Added General Information.

Date	12-Feb-2021
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-397 (/qs3/pubsys2/xml/en/procedures/377/377-019-397.html)
Procedure Title	Fuel Lift Pump Circuit
Update Reason	Added Cable, Electrical Breakout Harness, Part Number 3163531 to Additional Service Items list. Reorganized Resistance Check steps. Rewrote Voltage Check steps.

Date	04-Mar-2021
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	377-002-048 (/qs3/pubsys2/xml/en/procedures/377/377-002-048.html)
Procedure Title	Camshaft Cover Plate
Update Reason	Added procedure to manual.

Date	10-Mar-2021
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-022 (/qs3/pubsys2/xml/en/procedures/377/377-011-022.html)
Procedure Title	EGR Valve
Update Reason	Added step in Install section that indicated the EGR valve would be routed to the wiring harness and the EGR outlet temperature sensor wiring harness from the EGR valve mounting bracket attachment points around the EGR valve. Added new graphic depicting this action.
Date	17-Mar-2021
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-058 (/qs3/pubsys2/xml/en/procedures/493/493-011-058.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit
Update Reason	Added Diesel Exhaust Fluid Dosing Unit Inlet Filter Kit to Select Service Tools list.
Date	22-Mar-2021
Section	Section V - Specifications
Procedure Number	377-018-017 (/qs3/pubsys2/xml/en/procedures/377/377-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Corrected listed specification for At No-Load Governed Speed to reflect appropriate values.
Date	23-Mar-2021
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-026 (/qs3/pubsys2/xml/en/procedures/377/377-006-026.html)
Procedure Title	Injector
Update Reason	Made minor edits to procedure.

Date	30-Mar-2021
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-010 (/qs3/pubsys2/xml/en/procedures/377/377-014-010.html)
Procedure Title	Crankcase Blowby, Measure
Update Reason	Corrected Initial Check and Measure steps found in procedure.
Date	08-Apr-2021
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-165 (/qs3/pubsys2/xml/en/procedures/493/493-011-165.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Voltage Drop Test
Update Reason	Graphic added to general information step of procedure and updated graphic of diagram to show the checking of pins.
Date	21-Apr-2021
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Added With Self-Calibrating and Without Self-Calibrating Actuator task titles in Install and Calibration steps. Updated Fault Codes.
Date	27-Apr-2021
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-056 (/qs3/pubsys2/xml/en/procedures/493/493-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Changed the tolerance for DEF quality on various platforms as it was incorrect.

Date	24-May-2021
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	377-012-109 (/qs3/pubsys2/xml/en/procedures/377/377-012-109.html)
Procedure Title	Air Compressor Inlet Tube
Update Reason	Added Loctite® 242, Part Number 3824040 to Select Service Tools list. Added instruction to utilize Loctite® 242, Part Number 3824040 within procedure.

Date	26-May-2021
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-025 (/qs3/pubsys2/xml/en/procedures/377/377-005-025.html)
Procedure Title	Fuel Pump Gear Pump
Update Reason	Updated tools. Added General Information step.

Date	26-May-2021
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Updated graphics. Added Finishing Steps step.

Date	26-May-2021
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-024 (/qs3/pubsys2/xml/en/procedures/377/377-006-024.html)
Procedure Title	Fuel Supply Lines
Update Reason	Updated General Information.

Date	27-May-2021
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Updated gear pump pressure test measurement. Removed note. Added unit conversions.

Date	15-Jun-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-079 (/qs3/pubsys2/xml/en/procedures/377/377-001-079.html)
Procedure Title	Gear Cover, Upper
Update Reason	Added Loctite® 242, Part Number 3824040 to Select Service Tools list. Added instruction to apply Loctite® 242, Part Number 3824040 to the capscrew threads and bottom three studs. Corrected torque value for capscrews and hex nuts listed in Install step.

Date	20-Jul-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-028 (/qs3/pubsys2/xml/en/procedures/377/377-001-028.html)
Procedure Title	Cylinder Liner
Update Reason	Corrected maximum specification listed for the Cylinder Liner Protrusion (around individual cylinder).

Date	20-Jul-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-033 (/qs3/pubsys2/xml/en/procedures/377/377-001-033.html)
Procedure Title	Gear Housing, Front
Update Reason	Added instruction in remove step to remove/discard capscrews 2 through 13 for front gear housing and gasket.

Date	03-Aug-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-023 (/qs3/pubsys2/xml/en/procedures/377/377-001-023.html)
Procedure Title	Crankshaft Seal, Front
Update Reason	Added mention of crankshaft seal replacer kit, Part Number 4918991, into procedure.
Date	03-Aug-2021
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-063 (/qs3/pubsys2/xml/en/procedures/493/493-011-063.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Override Test
Update Reason	Minor wording correction made to procedure.
Date	11-Aug-2021
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	377-012-109 (/qs3/pubsys2/xml/en/procedures/377/377-012-109.html)
Procedure Title	Air Compressor Inlet Tube
Update Reason	Added information for adapters with an o-ring.
Date	24-Aug-2021
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-144 (/qs3/pubsys2/xml/en/procedures/493/493-011-144.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing System Pressure Test
Update Reason	Added instruction to set the multimeter to mV DC and zero the multimeter pressure sensor.

Date	13-Sep-2021
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-027 (/qs3/pubsys2/xml/en/procedures/377/377-014-027.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Regeneration Analyzer
Update Reason	Revised Select Service Tools list. Removed Analyzing the Data step from the procedure.

Date	28-Sep-2021
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-032 (/qs3/pubsys2/xml/en/procedures/377/377-014-032.html)
Procedure Title	Air Handling Performance Test
Update Reason	Added procedure to manual.

Date	12-Nov-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-088 (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
Procedure Title	Engine Base Timing
Update Reason	Added instruction to rotate the valve camshaft gear mounting capscrew an additional 120 degrees in the Adjust step.

Date	18-Nov-2021
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-021 (/qs3/pubsys2/xml/en/procedures/377/377-006-021.html)
Procedure Title	Fuel Manifold (Drain)
Update Reason	Added reference to procedure 010-023.

Date	29-Nov-2021
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Corrected wording of note in procedure.
Date	01-Feb-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-018 (/qs3/pubsys2/xml/en/procedures/377/377-001-018.html)
Procedure Title	Crankshaft Gear, Front (Crankshaft Installed)
Update Reason	Added Crankshaft adapter and gear remover kit, Part Number 5572864, Magnetic base drill, Part Number 5572871 (60 Hz) or 5572872 (50 Hz), and Cutter bit Insert, Part Number 5572875 to Select Service Tools list. Added Remove step that uses magnetic base drill.
Date	01-Feb-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-017 (/qs3/pubsys2/xml/en/procedures/377/377-001-017.html)
Procedure Title	Crankshaft Adapter
Update Reason	Added necessary tools to Select Service Tools list. Added Remove steps that utilize a magnetic base drill.

Date	24-Feb-2022
Section	Section 17 - Miscellaneous - Group 17
Procedure Number	377-017-011 (/qs3/pubsys2/xml/en/procedures/377/377-017-011.html)
Procedure Title	Straight Thread Plug
Update Reason	Corrected procedure to include General Information section that addresses repair of threaded holes using different methods while highlighting strengths and weaknesses of each method.
Date	10-Mar-2022
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-041 (/qs3/pubsys2/xml/en/procedures/493/493-011-041.html)
Procedure Title	Aftertreatment Diesel Particulate Filter
Update Reason	Added notes. Updated instructions for DPF.
Date	10-Mar-2022
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-085 (/qs3/pubsys2/xml/en/procedures/493/493-011-085.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Restriction Test
Update Reason	Updated instructions for DPF servicing.
Date	11-Apr-2022
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Added information to procedure that notes turbocharger actuators are sold with a calibration software installed.

Date	25-Apr-2022
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-056 (/qs3/pubsys2/xml/en/procedures/377/377-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Updated procedure with correct engine prefix.
Date	27-Apr-2022
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-034 (/qs3/pubsys2/xml/en/procedures/377/377-014-034.html)
Procedure Title	J1939 Data Link Diagnostic Tool
Update Reason	Added procedure to manual.
Date	26-May-2022
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-037 (/qs3/pubsys2/xml/en/procedures/377/377-014-037.html)
Procedure Title	Aftertreatment Intake NOx Sensor Test
Update Reason	Added to manual.
Date	26-May-2022
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-036 (/qs3/pubsys2/xml/en/procedures/377/377-014-036.html)
Procedure Title	Aftertreatment Outlet NOx Sensor Test
Update Reason	Added to manual.
Date	27-Jul-2022
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-236 (/qs3/pubsys2/xml/en/procedures/377/377-005-236.html)
Procedure Title	Fuel System Diagnostics
Update Reason	Added High-Pressure Fuel Pump Supply Flow Test steps to procedure.

Date	03-Aug-2022
Section	Section 5 - Fuel System - Group 05
Procedure Number	377-005-227 (/qs3/pubsys2/xml/en/procedures/377/377-005-227.html)
Procedure Title	Fuel Pump Head(s)
Update Reason	Removed all mentions of barrel and plunger assembly throughout procedure. Removed Rebuild steps from procedure.
Date	11-Aug-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-017 (/qs3/pubsys2/xml/en/procedures/377/377-001-017.html)
Procedure Title	Crankshaft Adapter
Update Reason	Corrected specifications listed for feeler gauge in the final Install stepblock.
Date	29-Aug-2022
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	493-019-478 (/qs3/pubsys2/xml/en/procedures/493/493-019-478.html)
Procedure Title	Aftertreatment Particulate Sensor
Update Reason	Removed electrical cable, Part Number 5299467 from procedure and replaced it with electrical Cable, Part Number 5394621.
Date	09-Sep-2022
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-036 (/qs3/pubsys2/xml/en/procedures/493/493-011-036.html)
Procedure Title	Aftertreatment Selective Catalytic Reduction (SCR) Catalyst
Update Reason	Minor correction renumbering SCR components listed in second Install step so that the text matches the accompanying graphic.

Date	23-Sep-2022
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-027 (/qs3/pubsys2/xml/en/procedures/377/377-010-027.html)
Procedure Title	Charge Air Cooler
Update Reason	Updated charge air cooler procedure to use pressure gauge instead of manometer.
Date	26-Sep-2022
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	377-007-031 (/qs3/pubsys2/xml/en/procedures/377/377-007-031.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Corrected the order in which graphics and instructions were shown in the procedure.
Date	27-Sep-2022
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	377-002-026 (/qs3/pubsys2/xml/en/procedures/377/377-002-026.html)
Procedure Title	Overhead Camshaft Bushings, Valve
Update Reason	Corrected minimum/maximum specifications listed for the valve camshaft bushings inside diameter.
Date	27-Sep-2022
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-036 (/qs3/pubsys2/xml/en/procedures/377/377-014-036.html)
Procedure Title	Aftertreatment Outlet NOx Sensor Test
Update Reason	Added that failure to perform these steps as they are listed in the procedure will result in false failure of the test.

Date	27-Sep-2022
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-037 (/qs3/pubsys2/xml/en/procedures/377/377-014-037.html)
Procedure Title	Aftertreatment Intake NOx Sensor Test
Update Reason	Added that failure to perform these steps as they are listed in the procedure will result in false failure of the test.
Date	07-Oct-2022
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-165 (/qs3/pubsys2/xml/en/procedures/493/493-011-165.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Voltage Drop Test
Update Reason	Corrected graphic of circuit.
Date	10-Oct-2022
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-017 (/qs3/pubsys2/xml/en/procedures/377/377-019-017.html)
Procedure Title	Engine Coolant Level Sensor
Update Reason	Added General Information section to procedure.
Date	28-Oct-2022
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-024-tr (/qs3/pubsys2/xml/en/procedures/377/377-010-024-tr.html)
Procedure Title	Air Leaks, Air Intake and Exhaust Systems
Update Reason	Corrected Select Service Tools list, added instructions for Engine and Air Compressor Air Intake Leak Test.

Date	02-Nov-2022
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	377-016-005 (/qs3/pubsys2/xml/en/procedures/377/377-016-005.html)
Procedure Title	Flywheel
Update Reason	Updated flywheel mounting capscrews torque specifications.

Date	02-Nov-2022
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	377-016-006 (/qs3/pubsys2/xml/en/procedures/377/377-016-006.html)
Procedure Title	Flywheel Housing
Update Reason	Flywheel housing bore diameter specifications corrected.

Date	10-Nov-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-112 (/qs3/pubsys2/xml/en/procedures/377/377-001-112.html)
Procedure Title	Camshaft Tone Wheel
Update Reason	Updated install step and corrected torque sequence.

Date	14-Nov-2022
Section	Section 8 - Cooling System - Group 08
Procedure Number	377-008-030 (/qs3/pubsys2/xml/en/procedures/377/377-008-030.html)
Procedure Title	Fan Drive Idler Pulley Assembly
Update Reason	Added procedure to manual.

Date	15-Nov-2022
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-144 (/qs3/pubsys2/xml/en/procedures/493/493-011-144.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing System Pressure Test
Update Reason	Removed mention of Part Number 5394563 and replaced it with Part Number 5299132.

Date	05-Dec-2022
Section	Section 14 - Engine Testing - Group 14
Procedure Number	493-014-017 (/qs3/pubsys2/xml/en/procedures/493/493-014-017.html)
Procedure Title	Snap Acceleration Test
Update Reason	Updated procedure to note that puffs of white and blue smoke when the engine is decelerating are expected, and does not indicate a failure.

Date	05-Dec-2022
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	377-016-005 (/qs3/pubsys2/xml/en/procedures/377/377-016-005.html)
Procedure Title	Flywheel
Update Reason	Corrected torque sequence listed in third step of Install section.

Date	13-Jan-2023
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-058 (/qs3/pubsys2/xml/en/procedures/377/377-011-058.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit
Update Reason	Clarified verbiage in regards to the dosing unit inlet screen best practices.

Date	13-Jan-2023
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-059 (/qs3/pubsys2/xml/en/procedures/377/377-011-059.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Valve
Update Reason	Added DEF Doser Valve leakage specifications.

Date	30-Jan-2023
Section	Section V - Specifications
Procedure Number	377-018-024 (/qs3/pubsys2/xml/en/procedures/377/377-018-024.html)
Procedure Title	Filter Specifications
Update Reason	Updated pressure side fuel filter part number.

Date	30-Jan-2023
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-065-tr (/qs3/pubsys2/xml/en/procedures/377/377-006-065-tr.html)
Procedure Title	Pressure Fuel Filter
Update Reason	Updated procedure to include the correct pressure side fuel filter micron size.

Date	30-Jan-2023
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	377-006-066-tr (/qs3/pubsys2/xml/en/procedures/377/377-006-066-tr.html)
Procedure Title	Fuel Filter Suction
Update Reason	Updating procedure to include the correct pressure side fuel filter micron size.

Date	10-Feb-2023
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-033 (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
Procedure Title	Turbocharger
Update Reason	Added metal debris to oil check details in Initial Check step.

Date	02-Mar-2023
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-056 (/qs3/pubsys2/xml/en/procedures/377/377-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Corrected exhaust gas pressure, intake manifold pressure, and turbocharger speed for a stationary (parked) regeneration.

Date	11-Apr-2023
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-152 (/qs3/pubsys2/xml/en/procedures/377/377-010-152.html)
Procedure Title	Variable Geometry Turbocharger Hysteresis Test
Update Reason	Added note to procedure to indicate Fault code 2198 can become active during the Variable Geometry Turbocharger Hysteresis Test.

Date	02-May-2023
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Restructured procedure. Added reference to service bulletin and training video.

Date	08-May-2023
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-165 (/qs3/pubsys2/xml/en/procedures/493/493-011-165.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Circuit Load Test
Update Reason	Added mention of Electronic Specialties 180 LOADpro® Dynamic Test Leads to procedure. Renamed procedure to Aftertreatment Diesel Exhaust Fluid Dosing Unit Circuit Load Test.
Date	15-May-2023
Section	Section 10 - Air Intake System - Group 10
Procedure Number	377-010-134 (/qs3/pubsys2/xml/en/procedures/377/377-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Added General Information section to procedure.
Date	20-Jun-2023
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-088 (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
Procedure Title	Engine Base Timing
Update Reason	Added task titles to the Adjust step for the two types of fuel pump idler gears, a few spelling corrections made as well.
Date	01-Aug-2023
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-056 (/qs3/pubsys2/xml/en/procedures/377/377-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Updated specifications listed for a stationary (parked) regeneration.

Date	21-Aug-2023
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-019-363 (/qs3/pubsys2/xml/en/procedures/377/377-019-363.html)
Procedure Title	Camshaft Position Sensor
Update Reason	Corrected typo found in procedure.
Date	05-Sep-2023
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-014 (/qs3/pubsys2/xml/en/procedures/377/377-014-014.html)
Procedure Title	Fuel Injector Performance Test
Update Reason	Removed statement indicating the timing associated with the fuel injector performance test.
Date	07-Sep-2023
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-037 (/qs3/pubsys2/xml/en/procedures/377/377-014-037.html)
Procedure Title	Aftertreatment Intake NOx Sensor Test
Update Reason	Removed statement indicating the timing associated with aftertreatment intake NOx sensor functional test.
Date	07-Sep-2023
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-036 (/qs3/pubsys2/xml/en/procedures/377/377-014-036.html)
Procedure Title	Aftertreatment Outlet NOx Sensor Test
Update Reason	Removed statement indicating the timing associated with the aftertreatment outlet NOx sensor functional test.

Date	07-Sep-2023
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-032 (/qs3/pubsys2/xml/en/procedures/377/377-014-032.html)
Procedure Title	Air Handling Performance Test
Update Reason	Removed statement indicating the timing associated with the test.

Date	07-Sep-2023
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-027 (/qs3/pubsys2/xml/en/procedures/377/377-014-027.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Regeneration Analyzer
Update Reason	Removed statement indicating the timing associated with the DPF regeneration analyzer test.

Date	30-Oct-2023
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	377-016-005 (/qs3/pubsys2/xml/en/procedures/377/377-016-005.html)
Procedure Title	Flywheel
Update Reason	Added reference to Service Bulletin 6506608 into procedure.

Date	09-Jan-2024
Section	Section 11 - Exhaust System - Group 11
Procedure Number	377-011-121 (/qs3/pubsys2/xml/en/procedures/377/377-011-121.html)
Procedure Title	Aftertreatment Diesel Exhaust Fluid Dosing Unit Suction Test
Update Reason	Dosing unit vaccum maximum specification corrected.

Date	08-Apr-2024
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	493-019-443 (/qs3/pubsys2/xml/en/procedures/493/493-019-443.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Differential/Outlet Pressure Sensor
Update Reason	Removed road test from finishing steps.
Date	06-Sep-2024
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-039 (/qs3/pubsys2/xml/en/procedures/377/377-014-039.html)
Procedure Title	Aftertreatment Diesel Particulate Filter High Idle Test
Update Reason	Added procedure to manual.
Date	09-Sep-2024
Section	Section V - Maintenance Specifications
Procedure Number	377-018-017 (/qs3/pubsys2/xml/en/procedures/377/377-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Updated specification format to match current standard. Removed past information regarding lubricating oil pan and lubricating oil system. Added updated details and specifications for various types of lubricating oil pans and the lubricating oil system.
Date	22-Nov-2024
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-034 (/qs3/pubsys2/xml/en/procedures/377/377-014-034.html)
Procedure Title	J1939 Data Link Diagnostic Tool
Update Reason	Removed references to the CAN Bus Diagnostics Tool and updated with approved J1939 Data Link Diagnostic Tool naming. Updated illustrations throughout the procedure to match change in tool name.

Date	27-Mar-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-036 (/qs3/pubsys2/xml/en/procedures/377/377-014-036.html)
Procedure Title	Aftertreatment Outlet NOx Sensor Test
Update Reason	Added details and instruction related to exhaust system purging into procedure.

Date	27-Mar-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-037 (/qs3/pubsys2/xml/en/procedures/377/377-014-037.html)
Procedure Title	Aftertreatment Intake NOx Sensor Test
Update Reason	Added details and instruction related to exhaust system purging into procedure.

Date	22-Apr-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-036 (/qs3/pubsys2/xml/en/procedures/377/377-014-036.html)
Procedure Title	Aftertreatment Outlet NOx Sensor Test
Update Reason	Updated Exhaust System Purge information in procedure.

Date	22-Apr-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-037 (/qs3/pubsys2/xml/en/procedures/377/377-014-037.html)
Procedure Title	Aftertreatment Intake NOx Sensor Test
Update Reason	Updated Exhaust System Purge information in procedure.

Date	08-May-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-032 (/qs3/pubsys2/xml/en/procedures/377/377-014-032.html)
Procedure Title	Air Handling Performance Test
Update Reason	Updated System Requirement section. Corrected Preparatory Steps to include additional information.
Date	19-Jun-2025
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	377-022-001_19 (/qs3/pubsys2/xml/en/procedures/377/377-022-001_19.html)
Procedure Title	Service Tools
Update Reason	Added 5299465 part number and description to tools list.
Date	08-Jul-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-039 (/qs3/pubsys2/xml/en/procedures/377/377-014-039.html)
Procedure Title	Aftertreatment Diesel Particulate Filter High Idle Test
Update Reason	Adding bullet to not continue with published troubleshooting if soot load is less than the specified value according to the platform and Soot-load based fault codes are inactive.
Date	19-Aug-2025
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	377-016-004 (/qs3/pubsys2/xml/en/procedures/377/377-016-004.html)
Procedure Title	Flexplate
Update Reason	Updated torque value for flexplate capscrews. Added information that some flexplates require mounting plates and/or clamp rings.

Date	26-Aug-2025
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-036 (/qs3/pubsys2/xml/en/procedures/493/493-011-036.html)
Procedure Title	Aftertreatment Selective Catalytic Reduction (SCR) Catalyst
Update Reason	Added reference to Service Bulletin, California Aftertreatment Regulation Adherence, Bulletin 6642787.

Date	26-Aug-2025
Section	Section 11 - Exhaust System - Group 11
Procedure Number	493-011-049 (/qs3/pubsys2/xml/en/procedures/493/493-011-049.html)
Procedure Title	Aftertreatment Diesel Oxidation Catalyst
Update Reason	Added reference to Service Bulletin, California Aftertreatment Regulation Adherence, Bulletin 6642787.

Date	03-Sep-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-014 (/qs3/pubsys2/xml/en/procedures/377/377-014-014.html)
Procedure Title	Fuel Injector Performance Test
Update Reason	Removed all ECM replacement callouts and replace them with direction to escalate to care.

Date	05-Sep-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-028 (/qs3/pubsys2/xml/en/procedures/377/377-014-028.html)
Procedure Title	Engine Compression and Air Handling System Test
Update Reason	Troubleshooting was added for symptoms will not complete, will not start, and did not pass. Removed tables from procedure and made minor edits for clarity.

Date	24-Sep-2025
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-025 (/qs3/pubsys2/xml/en/procedures/377/377-014-025.html)
Procedure Title	Aftertreatment Selective Catalytic Reduction (SCR) System Test
Update Reason	Updated procedure to current standards.
Date	10-Oct-2025
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-079 (/qs3/pubsys2/xml/en/procedures/377/377-001-079.html)
Procedure Title	Gear Cover, Upper
Update Reason	Added removal and installation of vibration damper to Preparatory and Finishing Steps of procedure.
Date	24-Nov-2025
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	377-001-047 (/qs3/pubsys2/xml/en/procedures/377/377-001-047.html)
Procedure Title	Piston Rings
Update Reason	Added Preparatory and Finishing Steps to procedure.
Date	24-Mar-2026
Section	Section 14 - Engine Testing - Group 14
Procedure Number	377-014-039 (/qs3/pubsys2/xml/en/procedures/377/377-014-039.html)
Procedure Title	Aftertreatment Diesel Particulate Filter High Idle Test
Update Reason	Removed statement that the engine speed needs to be increased in a certain RPM range. Removed note from procedure that stated if the engine speed can not be reached use Setup for Dynamometer in the Cummins® electronic service tool or equivalent and enable test throttle.

Date	19-Jun-2026
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	377-002-014 (/qs3/pubsys2/xml/en/procedures/377/377-002-014.html)
Procedure Title	Injector Sleeve, Cylinder Head
Update Reason	Removed mention of using a shim with the installation tool, adding instruction to use a dead blow hammer if needed.

Last Modified: 29-Jun-2026
